

FAHED AKIL SHAIKH

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OBJECTIVE

Electronics and Telecommunication Engineering graduate with hands-on experience in embedded C, ATmega328P, PIC18F4580, ARM, and Linux system programming, along with communication protocols including UART, SPI, I2C, and CAN. Built 9+ production-grade embedded and systems projects covering IPC, socket programming, custom shell implementation, and data structures, with additional exposure to IoT, cloud connectivity, and mobile-app-based monitoring systems.

SKILLS

Programming: C, C++, Python, Shell Scripting, Data Structures

Embedded Systems: GPIO, UART, SPI, I2C, CAN, Timers, Interrupts, EEPROM, Embedded C, RTOS Concepts, U-Boot

Dev Boards: PIC18F4580 (Microchip), ARM Cortex-M4 (STM32F446RE, STMicroelectronics), ATmega328P (Arduino), Raspberry Pi

Linux & Systems: Linux kernel system calls, IPC mechanisms - Pipe, FIFO, Shared Memory, Network programming using TCP and UDP sockets, Multi-threading, Process & Memory Management

Tools: Git, GCC, GDB, VS Code, Arduino IDE

PROFESSIONAL TRAINING

Embedded Systems Trainee

2025 - 2026

Emertxe Information Technologies, Bengaluru, Karnataka

Modules: LS, Advance C, C++, LI, DS, MC, ELARM, QT

- Built **11+ C/embedded projects** — including a custom **Mini Shell** with **process management**, **signal handling**, and **job control** — covering **IPC**, **socket programming**, and **data structures**.

PROJECTS

Car Black Box

(PIC18F4580, CAN Bus (MCP2551), I2C, RTC (DS1307), EEPROM (AT24C02), CLCD)

- Designed an automotive event data recorder (black box) on **PIC18F4580** that logs **gear shifts** with speed, **engine temperature**, **fuel consumption**, and **trip distance** in real time.
- Built a **password-protected menu system** (login, view log, set time) on a **CLCD**, with log entries timestamped using a **DS1307 RTC** and stored in an **AT24C02 EEPROM** over **I2C**.
- Interfaced an **MCP2551 CAN transceiver** for vehicle event data and implemented **GPIO polling**, **interrupts**, and **rollover-based log scrolling**.

CAN-Based Automotive Dashboard

(PIC18F4580, CAN Bus, Embedded C, CLCD, SSD, LED)

- Designed a **3-node CAN bus network**: Node 1 for speed and gear input, Node 2 for **RPM indicators**, and Node 3 for **CLCD, SSD, and LED display**.
- Implemented the **CAN communication protocol** for **real-time inter-node data exchange** across all three microcontroller boards.

Arbitrary Precision Calculator (APC)

(C, Linked Lists, Dynamic Memory Allocation)

- Performed arithmetic on integers beyond **64-bit limits** using **linked list storage** with optimized **carry/borrow handling**.

Inverted Search Engine

(C, Hash Tables, Linked Lists, File I/O)

- Developed a keyword-based **multi-file indexing system** using **hash tables and linked lists** for fast search and retrieval.

Trivial File Transfer Protocol (TFTP)

(C, UDP Sockets, Networking, Error Handling)

- Built a full **TFTP implementation** in C over **UDP** with **packet sequencing**, **ACK handling**, **timeout recovery**, and error scenarios.

Mini Shell (msh)

(C, fork/exec, Signals, Pipes, POSIX, Job Control)

- Built a **Linux shell** from scratch with command execution, **built-ins** (cd, pwd, exit), **PS1 customization**, and **special variables** (\$?, \$\$, \$SHELL).
- Implemented **signal handling** for **SIGINT/SIGTSTP** and full **job control** with fg, bg, jobs, background execution (&), and **SIGCHLD cleanup**.
- Supported **multi-stage pipes** (e.g. ls -l | grep tty | wc -l) with **dynamically created child processes**.

Source to HTML Converter

(C, File Handling, String Parsing)

- Built a C-based tool that converts C source files into **syntax-highlighted HTML** for readable, shareable code documentation.
- Implemented **tokenization and pattern matching** to identify keywords, comments, and strings for highlighting.

Lexical Analyzer

(C, String Parsing, File I/O, Token Classification)

- Tokenized C source files at the **character level**, classifying keywords, identifiers, operators, and literals into a **structured token stream**.

MP3 Tag Reader & Editor

(C, ID3 Tags, Binary File I/O)

- Developed a utility in C to read and edit **ID3v1/ID3v2 metadata tags** (title, artist, album, genre) embedded in MP3 files.
- Implemented **low-level binary file parsing** to locate and modify tag fields without corrupting audio data.

Image Steganography (LSB)

(C, Bit Manipulation, Binary File I/O)

- Hid and extracted secret text in image files using **LSB encoding**, demonstrating **bit-level manipulation** and binary file handling.

Address Book

(C, File Handling, Structures, Linked Lists)

- Built a **command-line contact management application** in C supporting add, edit, delete, and search operations.
- Implemented **persistent storage** using file handling so contact records are retained across sessions.

ACADEMIC PROJECTS

Raspberry Pi Farming Robot

(Raspberry Pi Pico W, Python, Image Processing, Cloud, IoT, Mobile App)

- Built an **autonomous robot** that captures soil and plant images, applies **image processing** for health analysis, and uploads sensor data to the **cloud**.
- Developed a **companion mobile app** for **real-time alerts** and live plant status monitoring.

Auto Temperature Detector - COVID Safety

(ATmega328P, IR Temp Sensor, Laser Diode, Bluetooth HC-05, LCD)

- Built an entrance safety system where a **laser diode** detects a person, an **IR sensor** checks temperature, and a **DC motor** grants or denies entry.
- Enforced room occupancy limits, with **Bluetooth integration** allowing **remote configuration** of temperature threshold and capacity via a mobile app.

AWARDS & CERTIFICATES

- Certified **Embedded Software Engineer (ELE/Q1501)**, NSQF Level-5 — Electronics Sector Skills Council of India (Skill India), via Emertxe Information Technologies [\[Link\]](#)
- Student Spot Award - Emertxe Information Technologies, for maintaining **100% attendance** in the C Programming Module [\[Link\]](#)
- C Programming For Beginners - Master the C Language**, Udemy [\[Link\]](#)

EDUCATION

B.E. - Electronics & Telecommunication Engineering

2025 / CGPA: 7.19

Ajeenkya DY Patil University, Pune (SPPU)

Diploma - Electronics & Telecommunication Engineering

2021 / 85.27%

Government Polytechnic, Pune